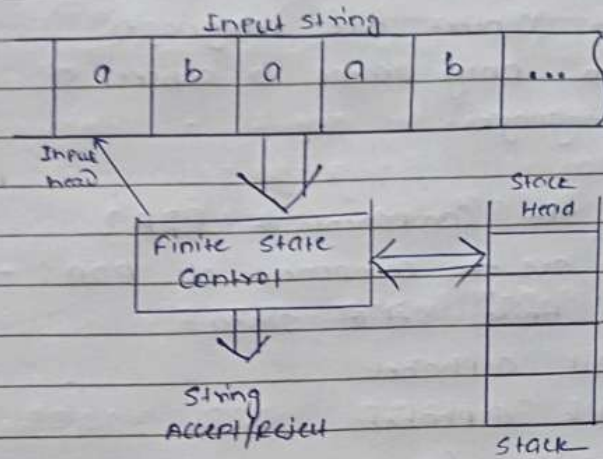


Section - A (2x5=10)

1. Explain block diagram of push down automata.

Ans



- Finite State Control :- Manage the state of the PDA.
- Stack: distinguishing feature of PDA, Provides additional memory.
- Input tape: contains the string to be processed by the PDA.

2. What are the two approaches of lang. acceptance in PDA?

Ans Two approaches are:-

- Acceptance by final state
- Acceptance by empty stack.

3. Define Instantaneous description (ID) in PDA.

Ans Instantaneous description (ID) is an informal notation of how a PDA computes an input string and makes a decision that string is accepted or rejected.

4. Is NPDA (Non deterministic PDA) an DPDA (Deterministic PDA) equivalent? Explain.

Ans NO. NPDA and DPDA are not equivalent in terms of expressive power.

- NPDA can accept broader class of languages, specifically context free languages.
- DPDA can only accept a subset of context-free languages.

Q. What are the components of PDA?

Ans following are the components of PDA: -

- Q : the finite set of states.
- Σ : input alphabets
- Γ : stack alphabet
- δ : transition fun.
- q_0 : initial state.
- Z_0 : initial stack symbol
- F : Set of accepting states.

• Section - B ($7 \times 3 = 21$)

1. Construct a PDA equivalent to the following CFG:

$S \rightarrow OBB, B \rightarrow OS/1S/0$. Test whether 010^4 is in $N(A)$

Ans $A = \{ (q), (0, 1), (S, B, 0, 1), S, q, S, \emptyset \}$

The production rule can be:

$$R_1: \delta(q, \epsilon, S) = \{ (q, OBB) \}$$

$$R_2: \delta(q, \epsilon, B) = \{ (q, OS), (q, 1S), (q, 0) \}$$

$$R_3: \delta(q, 1, 1) = \{ (q, \epsilon) \}$$

$$R_4: \delta(q, 0, 0) = \{ (q, \epsilon) \}$$

Analyzing 010^4 i.e. 010000 against a PDA

$$\delta(q, 010000, S) \rightarrow \delta(q, 010000, OBB)$$

$$\rightarrow \delta(q, 10000, BB) \quad R_1$$

$$\rightarrow \delta(q, 10000, 1SB) \quad R_2$$

$$\rightarrow \delta(q, 0000, SB) \quad R_3$$

$\rightarrow \delta(q, 0000, 0BBB) \quad R_1$
 $\rightarrow \delta(q, 000, BBB) \quad R_4$
 $\rightarrow \delta(q, 000, 0BB) \quad R_2$
 $\rightarrow \delta(q, 00, BB) \quad R_4$
 $\rightarrow \delta(q, 00, 0B) \quad R_2$
 $\rightarrow \delta(q, 0, B) \quad R_4$
 $\rightarrow \delta(q, 0, 0) \quad R_2$
 $\rightarrow \delta(q, \epsilon) \quad R_4$

Accept

Thus, 010^4 is accepted.

2. Explain PDA and its operation. Construct the PDA for the following languages: $\{1^n 2^m 3^n \mid n \geq 1, m \geq 1\}$

Ans PDA is a computational model used to recognize context free language, which include languages generated by context free grammars.

PDA operations:-

- (i) PUSH:- Adds a symbol to the top of stack
- (ii) POP:- Remove the top symbol from stack
- (iii) SKIP:- Leave the stacks unchanged but can read the I/P.

$$L = \{1^n 2^m 3^n \mid n \geq 1, m \geq 1\}$$

$$L = \{123, 11233, 112233, \dots\}$$

take input string $w: 11233$

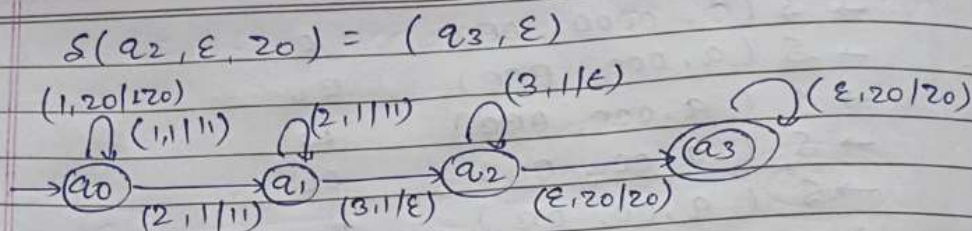
$$\delta(q_0, 1, 20) = (q_0, 120)$$

$$\delta(q_0, 1, 1) = (q_0, 11)$$

$$\delta(q_0, 2, 1) = (q_0, 11)$$

$$\delta(q_1, 3, 1) = (q_2, \epsilon)$$

$$\delta(q_2, 3, 1) = (q_3, \epsilon)$$



Q3

Explain PDA and its operation. Construct the PDA for the following language.

$$\{1^n 2^n 3^m \mid n \geq 1, m \geq 1\}$$

Ans

$$L = \{1^n 2^n 3^m \mid n \geq 1, m \geq 1\}$$

$$L = \{1, 2, 3, 1233, 11223, 112233, \dots\}$$

take input as: 112233

$$\delta(q_0, 1, z_0) = (q_0, 1z_0)$$

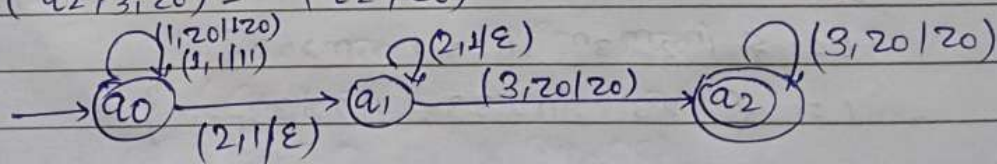
$$\delta(q_0, 1, 1) = (q_0, 11)$$

$$\delta(q_0, 2, 1) = (q_1, \epsilon)$$

$$\delta(q_1, 2, 1) = (q_1, \epsilon)$$

$$\delta(q_1, 3, z_0) = (q_2, z_0)$$

$$\delta(q_2, 3, z_0) = (q_2, z_0)$$



• Section-c (11x3=33)

1. Explain PDA and its operation. Construct the PDA for the following language.

$$\{1^n 2^{n+m} 3^m \mid n \geq 1, m \geq 1\}$$

Ans $L = \{1^n 2^{n+m} 3^m \mid n \geq 1, m \geq 1\}$

$L = \{1223, 11222233, 111222222333, 112223 \dots\}$

taking input strings: 112223 for $n=2, m=1$

$\delta(q_0, 1, z_0) \rightarrow (q_0 / 1z_0)$

$\delta(q_0, 1, 1) \rightarrow (q_0 / 11)$

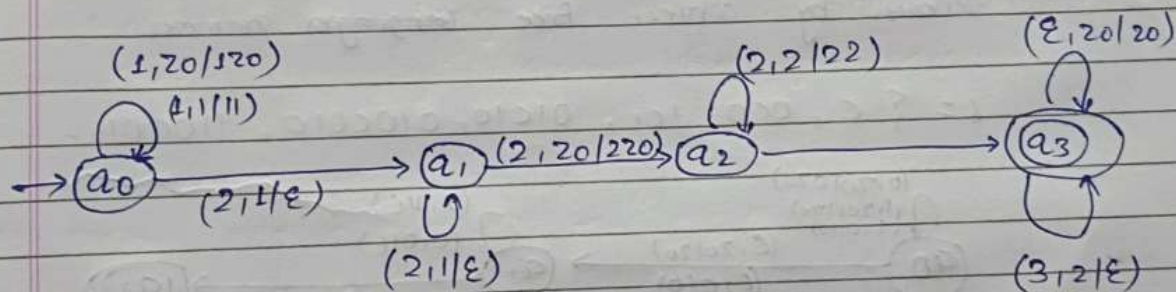
$\delta(q_0, 2, 1) \rightarrow (q_1 / \epsilon)$

$\delta(q_1, 2, 1) \rightarrow (q_1 / \epsilon)$

$\delta(q_1, 2, z_0) \rightarrow (q_2 / 2z_0)$

$\delta(q_2, 3, 2) \rightarrow (q_3 / \epsilon)$

$\delta(q_3, \epsilon, z_0) \rightarrow (q_3 / z_0)$



Q2 Explain PDA and its operation. Construct the PDA for the following languages: $\{wcw^R \mid w \in (0+1)^+\}$

Ans PDA is a computational model used to recognize context free language, which include language generated by context free grammars.

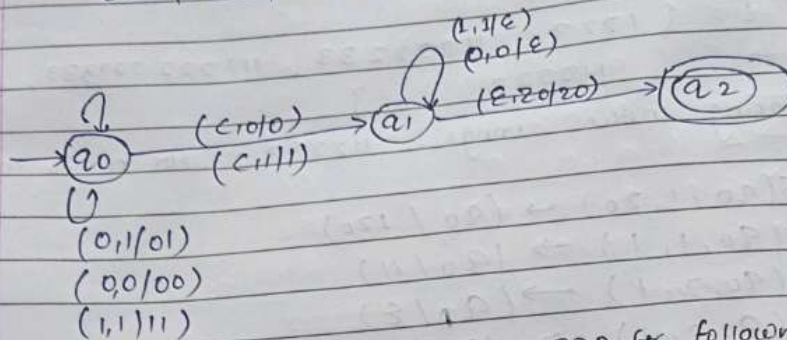
• PDA Operations: -

Push - Add the symbol

POP - Remove the symbol

skip - Leave the stack

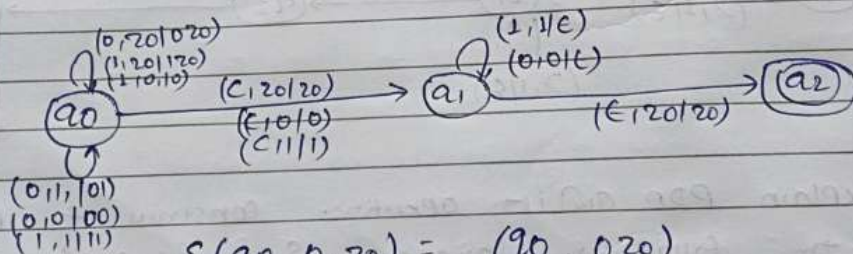
$$L = \{1c1, 0c0, 10c01, 01c10, 010c010, \dots\}$$



Q3 Explain PDA & its operation. Construct the PDA for following language
 $L = \{wcw^R \mid w \in \{0,1\}^*\}$

Ans PDA is a computational model used to recognize context free language, which include language generated by context free languages grammars.

$$L = \{c, 0c0, 1c1, 01c10, 010c010, 110c011, \dots\}$$



$$\begin{aligned}
 S(q_0, 0, 20) &= (q_0, 020) \\
 S(q_0, 1, 20) &= (q_0, 120) \\
 S(q_0, 0, 0) &= (q_0, 00) \\
 S(q_0, 0, 1) &= (q_0, 01) \\
 S(q_0, 1, 0) &= (q_0, 10) \\
 S(q_0, 1, 1) &= (q_0, 11) \\
 S(q_0, c, 0) &= (q_1, 0) \\
 S(q_0, c, 1) &= (q_1, 1) \\
 S(q_1, 0, 0) &= (q_1, \epsilon) \\
 S(q_1, 1, 1) &= (q_1, \epsilon) \\
 S(q_1, \epsilon, 20) &= (q_2, \epsilon)
 \end{aligned}$$